	In disease list	In 134 core	NPC log2FC	−log10 FDR	gutMGene records (supported/total)	Docked	SEA R0 metabolites	Best E46 R (total conc.)	GSE68799 log2FC (same dir., FDR<0.05)
MMP3	yes	yes	+3.28	10.1	1/1	no	1	0.045 (OK)	+0.05
MMP1	yes	yes	+3.27	11.2	—	no	1	79† (UNK)	+1.15
PTGS2	yes	yes	+2.45	9.7	4/5	yes	2	0.24 (OK)	+2.73 ✓
MMP12	no	no	+1.94	5.9	—	no	1	0.29 (OK)	+1.28
SOST	no	no	+1.59	4.6	—	no	1	no blood-conc. data	+0.51
TOP2A	yes	yes	+1.47	5.5	—	no	1	2.4e+02† (METHOD)	-0.21
CDK1	yes	yes	+1.31	7.5	—	no	1	0.57 (OK)	-0.05
CA9	yes	yes	+1.20	3.3	—	yes	7	11† (UNK)	+1.50 ✓
NOX4	no	no	+1.08	5.5	—	no	4	0.38 (OK)	+0.19
SLC1A3	no	no	+1.02	7.3	—	no	2	7.3 (PED)	+0.68
HSD17B2	no	no	-1.47	5.2	—	no	2	0.45 (OK)	+0.32
MAOB	yes	yes	-1.54	9.0	—	yes	3	1.5 (OK)	-0.88
AKR1C3	yes	yes	-2.04	8.2	—	no	1	1.8e+03 (MATRIX)	+0.71
ADRA2A	no	no	-2.09	10.8	—	yes	4	0.73 (PED)	-1.32

Scheme A: 14 robust (FDR<0.05, |log2FC|≥1, 3/3 datasets) SEA×STP targets (GPR84 left the set: its only SEA record was misattributed). R = blood upper bound / strongest activity, total concentration, source class from E63 (OK = adult baseline plasma; others flagged); † = flagged as doubtful concentration in E46; free-fraction correction not applied.

GSE68799 (RNA-seq, 42 NPC vs 4 controls; overlap with the three arrays not verified), Welch t-test with BH over all tested genes: 2/14 replicate, versus a baseline of 21% (195/916) for all robust DEGs, so no enrichment over baseline; 4 controls give limited power. Cell-type localisation (sample-level, selected genes) is in Fig. 9